

Data Cleaning Preparation

Introduction

Before starting analysis, I first cleaned the dataset because raw data usually contains errors, missing values, and duplicates. So that our ultimate goal of achieving valuable insight will not be possible. That's why I cleaned the dataset by all means.

This report describes the steps taken to clean and prepare the dataset.

Data Issues Found

While checking the data, I found:

- Remark columns had missing values but that doesn't affect any analysis
- Duplicate records were present in task but that doesn't affect any analysis
- Some columns had wrong data types
- There were some extreme values (outliers)

Handling Missing Values

- For numerical columns → I used **median**
- For categorical columns → I used **mode**

In some cases, I kept missing values if they had business meaning.

Removing Duplicates

Dataset given by client, does not contain any duplicated values.

Duplicate records were identified using pandas duplicated() function on valuable columns that affects to another column to avoid wrong analysis.

Data Type Fixing

- Converted date columns into datetime format
- Ensured numerical columns are in proper format (int / float)

Outlier

I used the **IQR method** to detect outliers. For that I used the pandas describe() function.

Adding Valuable Column

I created new useful column.

For example, for analyzing the any datetime column, I separated the Year column to analyze the year wise distribution of particular operation.

Final Dataset

After cleaning, the dataset became:

- More accurate
- Consistent
- Ready for analysis